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Athwart Loop

Implementation Plan v2 — Closing the gap to Handbook v5

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WHY A v2 — v1.0 (27 June 2026) planned a full rebuild from a 7-type/6-status legacy model. Since then, the team shipped the data model overhaul, the 3-status state machine, section-based routing, and full-text search — largely matching Handbook v5. This plan does not re-plan that work. It re-audits the live codebase against the handbook, part by part, and re-plans only what actually remains to close Phase 1 — reordered by what really blocks the pilot's Definition of Done, not by the original sequence.

METHOD NOTE — This audit read the actual backend and frontend source (schema, services, controllers, pages, stores) rather than relying on v1's now-stale description of "current state." File paths and line-level evidence are cited throughout so nothing here is a guess.

1. Already Shipped — Do Not Re-plan

The following v1 phases are done and match the handbook closely. Treat them as a stable foundation.

Area	Evidence	Matches Handbook
Post data model	schema.prisma Post model: postNumber, type (QUESTION/PROBLEM/IDEA), status (OPEN/DISCUSSING/RESOLVED), section (9 values), resolution (7 values), resolutionReason, isUseCase, ownerId, acknowledgedAt, linkedEntityType, linkedEntityId. priority and tags fully removed.	Yes — matches C1
State machine	workflow.service.ts: OPEN→DISCUSSING sets acknowledgedAt; →RESOLVED requires resolution + reason if PARKED/DECLINED; RESOLVED→OPEN reopen preserves acknowledgedAt; who-can-resolve enforced per type (Question: author or owner; Problem/Idea: owner only).	Yes — matches C2
Audit logging	audit.service.ts emits POST_ACKNOWLEDGED / POST_RESOLVED / POST_REOPENED on every transition.	Yes (USE_CASE_GRADUATED defined but unused — Phase 08)
Section routing	SectionOwnership model + seed-section-owners.ts + section.util.ts getSectionOwner() (falls back to FOUNDER); auto-assigned on POST /posts.	Yes — matches B7 / C2

Area	Evidence	Matches Handbook
Full-text search	Prisma fullTextSearchPostgres + GIN index migration (20260628211626_fts_index); feed-search.service.ts; ?search= param on GET /api/posts.	Yes, mechanism matches C4 (UI wiring gap below)

STACK NOTE (unchanged) — Still Express + Prisma + PostgreSQL, not Supabase/Lovable — the stack decision from v1 stands. Nothing in this plan revisits that.

2. Gap Analysis — What's Actually Missing

This replaces v1's gap analysis, which described a state that no longer exists. These are the real, current gaps.

Handbook requirement	Current state	Gap
CRM record linking (C3) — “Loop's highest-value feature”	Schema fields exist but are absent from the frontend Post type; composer has no CRM step; detail page has no CRM chip.	Fully unwired on the frontend — the single most valuable feature per the handbook is invisible to users.
AI E1 — search-first duplicate catcher	Not built. Only a generic “AI summary” feature exists (Groq workflow summary) — a different feature the handbook never specified.	Wrong AI feature built; E1 doesn't exist.
AI E2 — auto-draft for owners	Not built.	Doesn't exist.
Canonical answer block (D2)	Comments are a flat thread; nothing can be pinned or marked canonical.	Missing.
Resolution picker + reason UI (C2, D2)	Resolution shown as plain text; no picker or reason textarea on resolve.	Backend enforces resolution/reason correctly, but the UI gives users no correct way to satisfy it.
Owner action buttons (D2)	Generic status dropdown instead of “Start Discussing” / “Mark Resolved.”	UX doesn't reflect the state machine that already exists.
Use Case graduation (A2, D2)	isUseCase is only a checkbox at creation time.	Contradicts the handbook: graduation is meant to emerge from discussion, not be declared upfront.
Board default = Open (D1)	Feed loads with no status filter (shows everything).	Contradicts “the black-hole-prevention promise is the literal default view.”
Board status-pill counts (D1)	Shows a bare total count.	Missing the “N open · M need your answer” signal.
Kanban secondary toggle (D1)	Not built.	Missing.
My Posts filter (D1)	authorId/ownerId exist as generic params only, no one-click preset.	Missing “the single most important view for keeping people posting.”
SLA health + black-hole rate (C4)	Cron computes slaStatus hourly; no GET /sla-health, no black-hole metric.	Data exists, nothing is surfaced.
Contributions record (B10)	ProfilePage shows a raw recent-posts list.	Missing the factual “posts raised / resolved / answered” record the handbook specifies.

Handbook requirement	Current state	Gap
RBAC role taxonomy	Role enum is FRONTEND/BACKEND/DEVOPS/AI_ML — a software-team model.	Works today because SectionOwnership routes around it — a latent mismatch, not an active bug (see Phase 08).
Legacy `category` field	ArchivePage.tsx still filters on `category`, a field removed from Post entirely (backend/src/controllers etc. — confirmed at frontend/src/pages/ArchivePage.tsx:14,19).	Silently broken archive filtering, independent of everything else.

3. Phase Summary — Remaining Work

#	Phase	Effort	Priority
01	CRM Record Linking (End-to-End)	M	Blocking — highest handbook value
02	Post Lifecycle UI Completion	M	Blocking — Definition of Done
03	Board UI Completion	S	Blocking — Definition of Done
04	AI Layer E1 — Search-first Duplicate Catcher	M	Blocking — Definition of Done
05	AI Layer E2 — Auto-draft for Owners	M	Ship after 04 is stable
06	SLA Health Endpoint & Black-hole Rate	S	Phase 1
07	Contributions Record	S	Phase 1
08	RBAC Decision & Audit Cleanup	S	Phase 1
09	Legacy Cleanup (Archive `category` field)	XS	Do any time, no dependencies
—	Phase 2 — Voting, Weekly Digest, Campaigns, Full SLA Dashboard	—	Unchanged from v1 — correctly not started

Phase Detail

01 CRM Record Linking (End-to-End)

Effort M · Blocking

The handbook calls this “the multiplier” and Loop's highest-value feature (C3, Section 04). The backend schema already supports it — `linkedEntityType` / `linkedEntityId` exist on `Post`. This is now purely a frontend wiring gap.

IMPLEMENTATION TASKS

- Add `linkedEntityType` (BILL|CASE|PARTNER) and `linkedEntityId` to the frontend `Post` interface.
`frontend/src/types/index.ts`
- Add a CRM link step to the composer: entity-type selector + human-readable ref input (`bill_no` / `case_number` / `partner_id`), optional. `frontend/src/components/posts/CreatePostModal.tsx`

- Add a CRM link chip to the post detail header — e.g. “[BILL] SCCS0000000 →” — URL built from env vars (VITE_CRM_BILL_BASE_URL, VITE_CRM_CASE_BASE_URL, VITE_CRM_PARTNER_BASE_URL).
frontend/src/pages/PostDetailPage.tsx
- There is no live CRM to validate against in this standalone stack (unlike the handbook's Supabase-embedded assumption). Treat linkedEntityId as free text with light format validation only — do not build a picker that queries a real bills/cases/partners table that doesn't exist here. This is a deliberate, scoped-down adaptation of C3, not a shortcut to fix later.
- Surface the CRM chip on feed rows too (small icon + type badge), matching D1.
frontend/src/components/posts/PostCard.tsx

02 Post Lifecycle UI Completion

Effort M · Blocking

IMPLEMENTATION TASKS

- Replace the generic status dropdown with an inline status track plus explicit owner action buttons: “Start Discussing” (OPEN→DISCUSSING) and “Mark Resolved.” frontend/src/pages/PostDetailPage.tsx
- Add a resolution picker (options filtered by post type per C2) plus a reason textarea, required when Parked/Declined — this mirrors enforcement that already exists in the backend (backend/src/services/workflow.service.ts); the frontend just needs to collect it correctly.
- Add a canonical-answer block: owners mark a comment canonical; pin it, visually distinct, at the top of the thread. Requires a new isCanonical boolean on Comment. backend/prisma/schema.prisma, comment service, frontend/src/pages/PostDetailPage.tsx
- Add a “Graduated: Use Case” badge with an owner-only toggle (kebab menu), decoupled from the create-time checkbox — graduation is meant to emerge from discussion (per A2's clash example), not be declared upfront. Wire the existing but unused USE_CASE_GRADUATED audit event on toggle.
backend/src/controllers/post.controller.ts, backend/src/services/audit.service.ts

03 Board UI Completion

Effort S · Blocking

IMPLEMENTATION TASKS

- Default filter on load: status = OPEN. frontend/src/pages/FeedPage.tsx, frontend/src/stores/post.store.ts
- Header status-pill: “N open · M need your answer,” computed from counts already available client-side, or a small new counts endpoint if the full list isn't already fetched.
- Add a “My Posts” one-click filter preset (authorId = current user). frontend/src/stores/post.store.ts, FeedPage.tsx
- Add a secondary Kanban view toggle (Open / Discussing / Resolved columns), persisted in localStorage.
frontend/src/pages/FeedPage.tsx
- Fix ArchivePage.tsx — remove references to the removed category field; filter on type/section instead.
frontend/src/pages/ArchivePage.tsx:14, 19 (tracked separately as Phase 09, but land it here since it's the same file family).

04 AI Layer E1 — Search-first Duplicate Catcher

Effort M · Blocking

Retrieval, not generation. Never invents an answer; if nothing matches, stays silent. Never blocks post creation.

IMPLEMENTATION TASKS

- Add `POST /api/intelligence/duplicate-check`. Accepts `{ title, body? }`. Runs the existing Postgres FTS (`feed-search.service.ts`) scoped to `status=RESOLVED`, re-ranks top matches with Groq.
`backend/src/controllers/intelligence.controller.ts`, `backend/src/services/intelligence/`
- Response: `{ found: bool, match?: { postNumber, title, resolution, canonicalAnswer, url } }`. Canonical-answer excerpt max 300 chars. If the endpoint fails or times out, the composer proceeds normally.
- Frontend: debounced (500ms, min 10 chars) call on title input in the composer; non-blocking, dismissible hint card — “This may already be answered — [LOOP-2026-0042]. [Read it →]”.
`frontend/src/components/posts/CreatePostModal.tsx`
- Keep the existing “AI summary” workflow feature as-is — it's additive and doesn't conflict with E1; just don't mistake it for satisfying this requirement, which it doesn't.

05 AI Layer E2 — Auto-draft for Owners

Effort M · Ship after 04 is stable

Non-negotiable guardrails: Claude/Groq cannot close the loop. Only a human marks an answer canonical. The draft appears only in the owner's edit box, clearly labelled, and cites sources. Never auto-posted.

IMPLEMENTATION TASKS

- Add `POST /api/intelligence/draft-response/:postId`, owner-only (ownerId matches, or ADMIN/FOUNDER).
`backend/src/services/intelligence/`
- Prompt with post title, body, linked CRM text, and top 3 relevant resolved posts; ask for a 2-4 sentence draft with source post numbers appended. Response: `{ draft, sources[], confidence: 'high'|'low'|'none' }` — never fabricate; `confidence='none'` returns `draft: null`.
- “Generate draft response” button in `PostDetailPage` for owners on OPEN/DISCUSSING Questions; pre-fills the comment box, labelled “AI draft — review before posting,” with source chips.

06 SLA Health Endpoint & Black-hole Rate

Effort S · Phase 1

IMPLEMENTATION TASKS

- Add `GET /api/posts/sla-health`: black-hole rate (share of Open posts past 48h with no acknowledgedAt), counts by `slaStatus`, average time-to-acknowledge. Admin/Founder only.
- Surface `slaStatus` as a board filter/sort option — today it's badge-only on `PostCard/PostDetailPage`.
`frontend/src/pages/FeedPage.tsx`
- Keep the existing hourly cron (`backend/src/cron/sla.cron.ts`) as-is — hourly is a strict improvement over the handbook's “nightly”; no change needed there.

07 Contributions Record

Effort S · Phase 1

No scores, no leaderboard — a factual record only, per the handbook's explicit anti-gamification stance (Section 10).

IMPLEMENTATION TASKS

- Add GET /api/users/:id/contributions: total posts raised, posts with resolution ANSWERED/FIXED/APPROVED/RULE_DECIDED, posts where the user commented on a post that became ANSWERED. Self + Admin/Founder access. backend/src/controllers/user.controller.ts
- Extend ProfilePage.tsx (not a new route, to match the existing IA) with three panels: "Posts I raised," "Posts I helped resolve," "Questions I answered." Factual only — no count badges styled as scores.

08 RBAC Decision & Audit Cleanup

Effort S · Phase 1

IMPLEMENTATION TASKS

- USE_CASE_GRADUATED wiring is now covered by Phase 02's graduation toggle — no separate work needed here beyond confirming it fires.
- Decision needed, not urgent: keep the current Role enum (ADMIN/FOUNDER/Frontend/Backend/DevOps/AI_ML) since SectionOwnership already does the real per-section routing, or introduce v1's proposed loop.post/loop.resolve capability model. **Recommendation: keep current roles.** They already work via SectionOwnership; a second parallel authorization model adds risk without fixing a live problem. Revisit only if a real permission bug surfaces.

09 Legacy Cleanup

Effort XS · Do any time

IMPLEMENTATION TASKS

- Remove category references in frontend/src/pages/ArchivePage.tsx (lines 14, 19) — already folded into Phase 03, listed here for tracking completeness.

4. Phase 2 — Unchanged from v1

Confirmed not started, and correctly so — none of these are needed until the founding-team pilot is stable.

Feature	Status	Notes
Voting	Not started	Reaction model exists (emoji thumbs-up with counts) but is not a real one-vote-per-user Vote model or a "Top voted" sort. Build the Vote model + toggle endpoint when Phase 2 opens.
Weekly Digest	Not started	No cron, no email/notification aggregation for it. Reuse existing Notification infrastructure when built.
Campaigns	Not started	No Campaign model or routes exist.
Full SLA Dashboard	Not started	Phase 06 above ships the basic health endpoint; the full admin dashboard with sparklines and per-owner performance is still Phase 2 scope.

5. Definition of Done — Phase 1, Scored Against Reality

Criterion	Status	Blocked on
Typed, section-tagged post	Done	—
Linked to a real CRM record	Not done	Phase 01
Owner moves Open → Discussing → Resolved	Backend done, frontend UX incomplete	Phase 02
Canonical answer	Not done	Phase 02
Transition is audit-logged	Done	—
Search-first AI hint before duplicate posting	Not done	Phase 04
Next person finds the answer via search	Done	—

BOTTOM LINE — Three of seven Definition-of-Done criteria are fully met, one is backend-complete but frontend-incomplete, and three are genuinely missing. None of the gap is in backend architecture — the state machine, routing, and search are solid. The remaining work is almost entirely UI (Phases 01–03) and the two AI features (Phases 04–05). That is good news: it means the riskiest, hardest-to-change layer is already done, and what's left is comparatively fast to build.